

FITBUDDY AI

4. Project Planning Phase

PROJECT REPORT DOCUMENT

Project: FitBuddy AI — Fitness Plan Generator

Group 1

M. Vijayasathy — Team Leader

Bharath

Harsha

Sumith

Arun Vasan

Objective

To organize project scope, deliverables, roles, schedule, resources, risks and milestones.

Prepared for academic project submission

4. Project Planning Phase

Scope

- In scope: profile form, validation, seven-day plan generation, wellness tip, feedback-based revision, database persistence, admin-protected user listing, REST API, testing and documentation.
- Out of scope: wearable-device integration, payment, live trainer consultation, medical diagnosis, clinical nutrition plans and production-scale deployment.

Work Breakdown & Schedule

- Stage 1 — Ideation and scope: define problem, audience and features.
- Stage 2 — Requirements: document inputs, outputs, constraints and acceptance criteria.
- Stage 3 — Design: architecture, data model, routes and UI flow.
- Stage 4 — Development: frontend templates, backend routes, AI service and database.
- Stage 5 — Testing: validation, API, feedback, access-control and error-path tests.
- Stage 6 — Documentation: technical setup, user guide and project report.
- Stage 7 — Demonstration: prepare sample profile, demonstrate generation and feedback revision.

Team Responsibilities

- M. Vijayasathy (Team Leader): coordination, task tracking and presentation integration.
- Bharath: requirements and documentation support.
- Harsha: UI/template and user-flow support.
- Sumith: backend/API and database support.
- Arun Vasan: testing, validation and demonstration support.
- These are proposed working allocations and can be adjusted to match actual contributions.

Resources

- Hardware: laptop/desktop and internet connection for live AI calls.
- Software: VS Code, Python 3.11+, virtual environment, browser, Git (optional).
- Libraries/services: FastAPI, Jinja2, SQLAlchemy, Pydantic, Google GenAI SDK, pytest and Gemini API access.

Milestones

- M1: approved concept and scope.
- M2: requirements and architecture finalized.
- M3: profile form and persistence working.
- M4: AI plan and tip generation working.
- M5: feedback revision and API completed.
- M6: tests and documentation completed.
- M7: demonstration-ready build.

Risk Register

- API key/model availability: provide mock mode and configurable model IDs.
- AI response formatting issues: validate/format output and show controlled errors.
- Unsafe or unsuitable suggestions: safety-focused prompts and clear wellness disclaimer; review outputs during demo.
- Credential exposure: use environment variables and replace default admin credentials.
- Schedule delays: prioritize core generation and feedback flow before optional polish.
- Data loss in local demo: retain database backup and use sample data.

Completion Criteria

- All core user flows operate locally, required tests pass, setup instructions are reproducible, safety limitations are stated and demo data is ready.

Project Team — Group 1

- M. Vijayasathya — Team Leader
- Bharath
- Harsha
- Sumith
- Arun Vasan